

The Mordell-Type Weierstrass Curve Associated to the Given Equation

Derived from the parameterized equation

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1. Compression of the original equation

Write the equation in the compact form

$$y^2 = (12x + 7 + L_1k + L_0)^2 + \frac{P(k)}{12x + 7}, \quad (1)$$

where

$$P(k) = C_3k^3 + C_2k^2 + C_1k + C_0.$$

The constants from the prompt are:

L_1	30619460690539143960110429669568004739580064102209388460790
L_0	10924398763302953739463149974094322610088557932235260987134
C_3	4784552901717657046910952386140410566083182302650026877054752121660964103096189843657017 751024472339276482878734850933847930096466248131898297321226278860780184291784358506500
C_2	5121092529754964411449161340146101420228399591452111284241547501882790784594208318617802 044125288709834578672750549563070264585752458229541411205828336479875946370066410764700
C_1	1827101315213597820000110416948106607208358810083338384275965340441733187215493526894277 304231141196649229390998742010658767143176530548275036317765383648969084915427119792620
C_0	2172908220045375678212962807621251486284512407003554783714681138381068170353585936805483 75733558724611766519425198024760434348671386193909776843642323732294542277989631353665

Both linear coefficients are divisible by 6. Define

$$n = n(k) = ak + b, \quad (2)$$

where

$$\begin{aligned} a &= L_1/6 \\ &5103243448423190660018404944928000789930010683701564743465 \\ b &= L_0/6 \\ &1820733127217158956577191662349053768348092988705876831189 \end{aligned}$$

Then

$$L_1k + L_0 = 6n(k). \quad (3)$$

A direct coefficient comparison gives

$$C_3 = 36a^3, \quad C_2 = 108a^2b, \quad C_1 = 108ab^2, \quad C_0 = 36b^3 - 19. \quad (4)$$

Indeed,

$$36(ak + b)^3 - 19 = 36a^3k^3 + 108a^2bk^2 + 108ab^2k + (36b^3 - 19).$$

Therefore the whole numerator is exactly

$$P(k) = 36n(k)^3 - 19. \quad (5)$$

Now put

$$q = 12x + 7. \quad (6)$$

Using (3) and (5), equation (1) becomes

$$y^2 = (q + 6n)^2 + \frac{36n^3 - 19}{q}, \quad n = n(k), \quad q = 12x + 7. \quad (7)$$

The denominator condition is $q \neq 0$, equivalently $12x + 7 \neq 0$.

2. Derivation of the Weierstrass curve

Let

$$R = 36n^3 - 19. \quad (8)$$

Then (7) is

$$y^2 = (q + 6n)^2 + \frac{R}{q}. \quad (9)$$

Define

$$U = \frac{R}{q}, \quad Y = Uy = \frac{Ry}{q}. \quad (10)$$

Since $q = R/U$, equation (9) gives

$$\begin{aligned} Y^2 &= U^2 y^2 \\ &= U^2 \left(\left(\frac{R}{U} + 6n \right)^2 + U \right) \\ &= (R + 6nU)^2 + U^3 \\ &= U^3 + 36n^2 U^2 + 12nRU + R^2. \end{aligned}$$

Thus the intermediate cubic is

$$Y^2 = U^3 + 36n^2 U^2 + 12nRU + R^2. \quad (11)$$

Remove the quadratic term by shifting

$$X = U + 12n^2, \quad U = X - 12n^2. \quad (12)$$

Substituting (12) into (11) gives

$$\begin{aligned} Y^2 &= (X - 12n^2)^3 + 36n^2(X - 12n^2)^2 \\ &\quad + 12nR(X - 12n^2) + R^2. \end{aligned}$$

Using $R = 36n^3 - 19$, the X^2 -coefficient cancels. The coefficient of X is

$$12nR - 432n^4 = 12n(36n^3 - 19) - 432n^4 = -228n.$$

The constant term is

$$\begin{aligned} 3456n^6 - 144n^3R + R^2 &= 3456n^6 - 144n^3(36n^3 - 19) + (36n^3 - 19)^2 \\ &= -432n^6 + 1368n^3 + 361. \end{aligned}$$

Therefore the associated short Weierstrass equation is

$$\boxed{Y^2 = X^3 - 228nX - 432n^6 + 1368n^3 + 361.} \quad (13)$$

Equivalently,

$$\boxed{Y^2 = X^3 - 228nX - (432n^6 - 1368n^3 - 361).} \quad (14)$$

3. Final curve attached to the parameter k

Substituting $n = n(k) = ak + b$ into (13), the curve attached to the original equation is

$$\boxed{E_k : Y^2 = X^3 - 228n(k)X - 432n(k)^6 + 1368n(k)^3 + 361.} \quad (15)$$

Equivalently,

$$\boxed{E_k : Y^2 = X^3 - 228n(k)X - (432n(k)^6 - 1368n(k)^3 - 361).} \quad (16)$$

Here

$$n(k) = ak + b,$$

with a and b given in Section 1.

This is the natural Mordell-type elliptic curve in short Weierstrass form. Strictly speaking, it is not a pure Mordell curve of the special form $Y^2 = X^3 + D$, because the linear term $-228n(k)X$ is present.

4. Birational correspondence

The forward change of variables from the original equation to the Weierstrass equation is

$$\boxed{X = 12n^2 + \frac{36n^3 - 19}{12x + 7}, \quad Y = \frac{(36n^3 - 19)y}{12x + 7}.} \quad (17)$$

Conversely, away from the exceptional denominator $X = 12n^2$, one recovers

$$\boxed{12x + 7 = \frac{36n^3 - 19}{X - 12n^2}, \quad y = \frac{Y}{X - 12n^2}.} \quad (18)$$

Equivalently,

$$\boxed{x = \frac{1}{12} \left(\frac{36n^3 - 19}{X - 12n^2} - 7 \right), \quad y = \frac{Y}{X - 12n^2}.} \quad (19)$$

Thus rational points on E_k with $X \neq 12n(k)^2$ correspond rationally to solutions of the original equation, subject to $12x + 7 \neq 0$.

For integer solutions in the original variables, the inverse map must additionally produce integral values. In particular,

$$q = \frac{36n^3 - 19}{X - 12n^2}$$

must be an integer satisfying

$$q \equiv 7 \pmod{12},$$

and $Y/(X - 12n^2)$ must be an integer.

5. Nonsingularity

For a short Weierstrass curve

$$Y^2 = X^3 + \mathcal{A}X + \mathcal{B},$$

we have

$$\mathcal{A} = -228n, \quad \mathcal{B} = -432n^6 + 1368n^3 + 361.$$

A direct factorization gives

$$4\mathcal{A}^3 + 27\mathcal{B}^2 = 27(4n^3 - 19)(36n^3 - 19)^3. \quad (20)$$

Therefore the discriminant is

$$\Delta = -16(4\mathcal{A}^3 + 27\mathcal{B}^2) = -432(4n^3 - 19)(36n^3 - 19)^3. \quad (21)$$

For every integer n , this is nonzero, since neither $4n^3 = 19$ nor $36n^3 = 19$ has an integer solution. Hence for integer k , the curve E_k is nonsingular over $\mathbb{Q}$.

6. Summary

The huge equation reduces to

$$y^2 = (q + 6n)^2 + \frac{36n^3 - 19}{q}, \quad q = 12x + 7, \quad n = n(k) = ak + b.$$

The associated curve is

$$\boxed{Y^2 = X^3 - 228nX - 432n^6 + 1368n^3 + 361,}$$

where $n = n(k)$. Equivalently,

$$\boxed{E_k : Y^2 = X^3 - 228n(k)X - (432n(k)^6 - 1368n(k)^3 - 361).}$$

The explicit correspondence is

$$X = 12n^2 + \frac{36n^3 - 19}{12x + 7}, \quad Y = \frac{(36n^3 - 19)y}{12x + 7}.$$